

We consider a finite nonempty multiset S of positive integers < 20 . Let

$$f(x) = \sum_{a \in S} x^a$$

be its generating function (coefficient of x^a equals the multiplicity of a). For a fixed positive integer i the condition that S can be partitioned into A_i, B_i with $A_i = i + B_i$ is equivalent to

$$f(x) = (1 + x^i)g_i(x)$$

for some polynomial $g_i(x)$ with non-negative integer coefficients. The index of S is the largest n for which this holds for every $i = 1, \dots, n$; if no such n exists the index is 0.

Let $L_n(x)$ be the least common multiple (in $\mathbb{Z}[x]$) of the polynomials $1 + x^i$ for $i = 1, \dots, n$. If S has index at least n then each $1 + x^i$ divides $f(x)$, hence $L_n(x)$ divides $f(x)$; therefore

$$f(x) = L_n(x)h(x)$$

for some polynomial $h(x)$ with integer coefficients. Moreover $f(0) = 0$ and $\deg f \leq 19$ (because all elements are < 20).

0.0.1 1. The maximum possible index N

The cyclotomic factorisation of $1 + x^i$ is

$$1 + x^i = \prod_{\substack{d|2i \\ d \nmid i}} \Phi_d(x),$$

where Φ_d are the cyclotomic polynomials. Hence

$$L_n(x) = \prod_{d \text{ primitive for some } i \leq n} \Phi_d(x).$$

For $n = 5$ the relevant divisors are 2, 4, 6, 8, 10; for $n = 6$ we also have 12. Thus

$$L_5(x) = \Phi_2\Phi_4\Phi_6\Phi_8\Phi_{10}, \quad L_6(x) = L_5(x)\Phi_{12}.$$

Their degrees are

$$\deg L_5 = 1 + 2 + 2 + 4 + 4 = 13, \quad \deg L_6 = 13 + 4 = 17.$$

Index 6 is impossible. Assume that a multiset S has index ≥ 6 . Then $f = L_6h$ with $\deg f \leq 19$, so $\deg h \leq 2$. Write $h(x) = ax + bx^2$ (a, b integers). Because $f(0) = 0$ we have $h(0) = 0$. Now consider the condition for $i = 1$. Set $Q_1(x) = L_6(x)/(1 + x)$. Using the product form,

$$Q_1(x) = (\Phi_4\Phi_6\Phi_8\Phi_{10})(1 - x^2 + x^4)$$

and an explicit multiplication gives

$$Q_1(x) = 1 - 2x + 3x^2 - 3x^3 + 4x^4 - 4x^5 + 5x^6 - 5x^7 + 6x^8 - 5x^9 + 5x^{10} - 4x^{11} + 4x^{12} - 3x^{13} + 3x^{14} - 2x^{15} + x^{16}.$$

The product $Q_1(x)h(x)$ must have non-negative coefficients. Computing the coefficients yields the inequalities

$$\begin{aligned}
k = 1 : a &\geq 0, \\
k = 2 : -2a + b &\geq 0 \Rightarrow b \geq 2a, \\
k = 3 : 3a - 2b &\geq 0 \Rightarrow b \leq \frac{3}{2}a, \\
k = 11 : 5a - 5b &\geq 0 \Rightarrow a \geq b, \\
k = 17 : a - 2b &\geq 0 \Rightarrow a \geq 2b.
\end{aligned}$$

The conditions $b \geq 2a$ and $b \leq \frac{3}{2}a$ force $b = 2a$; then $a \geq 2b$ gives $a \geq 4a$, so $a = 0$ and consequently $b = 0$. Hence no non-zero polynomial h of degree ≤ 2 exists, contradicting the existence of a non-empty S with index 6. Therefore $N \leq 5$.

Index 5 is attainable. Define

$$f(x) = x \prod_{i=1}^5 (1+x^i) = x + x^2 + x^3 + 2x^4 + 2x^5 + 3x^6 + 3x^7 + 3x^8 + 3x^9 + 3x^{10} + 2x^{11} + 2x^{12} + x^{13} + x^{14} + x^{15} + x^{16}.$$

For each $i = 1, \dots, 5$ we have

$$\frac{f(x)}{1+x^i} = x \prod_{\substack{j=1 \\ j \neq i}}^5 (1+x^j),$$

which is a product of polynomials with non-negative coefficients, hence has non-negative coefficients. Thus the index of this multiset is at least 5. Combined with $N \leq 5$ we obtain $N = 5$.

0.0.2 2. The smallest possible maximum element M for index 5

Let S have index 5 and write $f(x) = L_5(x)h(x)$ with $h(0) = 0$ and $\deg f \leq 19$. Because $\deg L_5 = 13$, we have $\deg h \leq 6$. We first show that $\deg h \geq 3$. Suppose $\deg h \leq 2$, write $h(x) = ax + bx^2$. Now use the condition for $i = 1$: $f/(1+x) = (L_5/(1+x))h$ must have non-negative coefficients. Set

$$Q_1^{(5)}(x) = \frac{L_5(x)}{1+x} = \Phi_4\Phi_6\Phi_8\Phi_{10} = 1 - 2x + 4x^2 - 5x^3 + 7x^4 - 7x^5 + 8x^6 - 7x^7 + 7x^8 - 5x^9 + 4x^{10} - 2x^{11} + x^{12}.$$

The product $(ax + bx^2)Q_1^{(5)}$ gives the inequalities

$$\begin{aligned}
x^2 : -2a + b &\geq 0 \Rightarrow b \geq 2a, \\
x^3 : 4a - 2b &\geq 0 \Rightarrow b \leq 2a,
\end{aligned}$$

so $b = 2a$. Then the coefficient of x^5 is $7a - 5b = 7a - 10a = -3a \geq 0$, which forces $a = 0$ and hence $b = 0$. Thus no non-zero h of degree ≤ 2 exists. Therefore $\deg h \geq 3$, and consequently $\deg f = \deg L_5 + \deg h \geq 13 + 3 = 16$. Hence the maximum element M of S satisfies $M \geq 16$.

The example above has maximum element 16; therefore $M = 16$.

0.0.3 3. The smallest possible number of elements for index 5 and maximum element 16

Now assume that S has index 5 and its maximum element is exactly 16. Then $\deg f = 16$ and $\deg h = 3$. Write $h(x) = ax + bx^2 + cx^3$ with integers a, b, c . Because all coefficients of f are

non-negative and $f(0) = 0$, we have $a, b, c \geq 0$ (the coefficient of x^1 is a , etc.). The number of elements of S is

$$f(1) = L_5(1)h(1) = 8(a + b + c),$$

since $L_5(1) = \Phi_2(1)\Phi_4(1)\Phi_6(1)\Phi_8(1)\Phi_{10}(1) = 2 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 8$. To minimise the number of elements we must minimise $a + b + c$ subject to the condition that for every $i = 1, \dots, 5$ the polynomial $Q_i(x)h(x)$ has non-negative coefficients, where $Q_i(x) = L_5(x)/(1 + x^i)$.

The most restrictive condition comes from $i = 1$. Using the same $Q_1^{(5)}(x)$ as above, the coefficients of $(ax + bx^2 + cx^3)Q_1^{(5)}$ must be non-negative. Computing these coefficients gives the following inequalities (the coefficient of x^k is $a q_{k-1} + b q_{k-2} + c q_{k-3}$, where q_j are the coefficients of $Q_1^{(5)}$):

k	coefficient
2	$-2a + b$
3	$4a - 2b + c$
4	$-5a + 4b - 2c$
5	$7a - 5b + 4c$
6	$-7a + 7b - 5c$
7	$8a - 7b + 7c$
8	$-7a + 8b - 7c$
9	$7a - 7b + 8c$
10	$-5a + 7b - 7c$
11	$4a - 5b + 7c$
12	$-2a + 4b - 5c$
13	$a - 2b + 4c$
14	$b - 2c$
15	c

All of these must be ≥ 0 . The first and last inequalities give $b \geq 2a$ and $b \geq 2c$, so $a, b, c \geq 0$.

Solving the system. If $a = 0$ then from $b \geq 2c$ and the inequality for $k = 5$ ($7 \cdot 0 + 4c \geq 5b$) we obtain $4c \geq 5b \geq 10c$, hence $c = 0$ and then $b = 0$. Thus $a \geq 1$.

For $a = 1$ we have $b \geq 2$. The inequality $b \geq 2c$ together with $a + 4c \geq 2b$ (from $k = 13$) gives

$$1 + 4c \geq 2b \geq 4c \implies 1 \geq 0,$$

which is always true. The strongest constraints are the ones from the coefficients of x^5 and x^6 :

$$7 + 4c \geq 5b, \quad 7 + 5c \leq 7b.$$

For $b = 2$ we obtain $4c \geq 3$ and $5c \leq 7$, so $c = 1$. For $b = 3$ we would need $c \geq ?$ but then $b \geq 2c$ forces $c \leq 1$, and the inequality $7 + 5c \leq 7b$ becomes $7 + 5c \leq 21$, which is satisfied, but the inequality $7 + 4c \geq 5b$ gives $7 + 4c \geq 15 \Rightarrow 4c \geq 8 \Rightarrow c \geq 2$, contradicting $c \leq 1$. Hence the only possibility for $a = 1$ is $(a, b, c) = (1, 2, 1)$.

For $a \geq 2$ we have $b \geq 2a \geq 4$. The inequality $b \geq 2c$ together with $a + 4c \geq 2b$ gives

$$2b \leq a + 4c \leq a + 2b \quad (\text{since } c \leq b/2),$$

which is automatically true. The inequality $7a + 5c \leq 7b$ forces $c \leq (7b - 7a)/5$. For $a = 2$ and $b = 4$ we get $c \leq (28 - 14)/5 = 3.2$, and from $b \geq 2c$ we have $c \leq 2$. The inequality $7a + 4c \geq 5b$ gives $14 + 4c \geq 20 \Rightarrow 4c \geq 6 \Rightarrow c \geq 2$. Hence $c = 2$. Then $a + b + c = 8$. For larger a the sum is even larger. Thus the smallest possible sum is $a + b + c = 4$, attained at $(1, 2, 1)$.

Therefore the number of elements is at least $8 \cdot 4 = 32$. The example $f(x) = x \prod_{i=1}^5 (1 + x^i)$ corresponds to $h(x) = x + 2x^2 + x^3$ and has exactly 32 elements. Hence the minimal possible number of elements is $m = 32$.

0.0.4 4. Conclusion

$$\boxed{N = 5}, \quad \boxed{M = 16}, \quad \boxed{m = 32}.$$